

SUPPLY CHAIN ANALYSIS REPORT

DataCo Global Dataset — Comprehensive Project Report

Dataset:	DataCo Supply Chain Dataset
Records Analyzed:	172,765 orders (post-cleaning)
Time Period:	January 2015 – January 2018
Tools Used:	Python Pandas Matplotlib Seaborn
Report Date:	May 2026

Executive Summary

This report presents a comprehensive data-driven analysis of the DataCo Global Supply Chain dataset, encompassing 172,765 order records spanning three years (2015–2018). The analysis covers exploratory data analysis, profitability assessment, delivery performance, bottleneck detection, root cause identification, and time-series trends.

The findings reveal a supply chain under significant delivery strain — over half of all orders experience late delivery. The financial impact is substantial: delayed orders account for \$2.1M in associated profit erosion. However, the business remains broadly profitable with 80.7% of orders generating positive returns.

Key Highlights

- 172,765 clean orders analyzed across 10 global regions
- Overall late delivery rate: 54.71% — a critical operational risk
- Total profit generated: \$7.5M | Profit at risk from delays: \$2.1M
- 80.7% of orders are profitable; 18.7% result in financial losses
- Standard Class shipping dominates (59.7%) yet has the lowest delay rate
- Central Africa is the highest-delay region at 61.3%
- First Class shipping in Central Africa shows a 100% delay rate
- Payment Review orders experience the highest delay rate (58%) by order status

Profitability Distribution

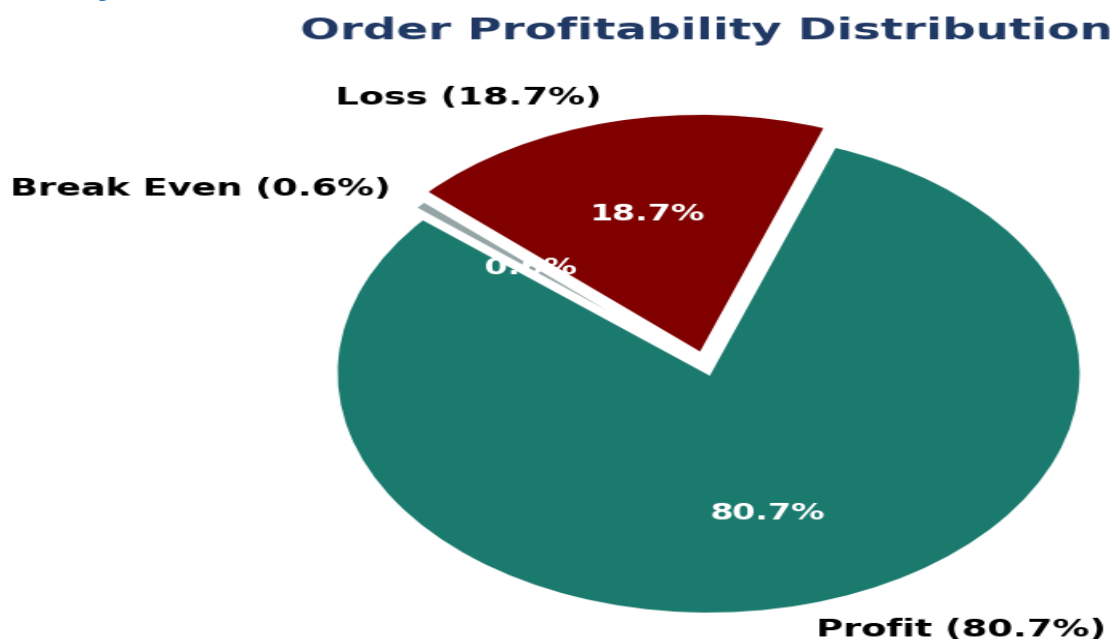


Figure 1 — Order Profitability Distribution: 80.7% Profitable | 18.7% Loss | 0.6% Break Even

Dataset Overview & EDA

The raw dataset had 180,519 rows and 53 columns. After removing cancelled shipments and PII/irrelevant columns, the working dataset was refined to 172,765 rows and 20 analytical columns.

Dataset Dimensions

Metric	Raw Dataset	Clean Dataset
Total Rows	180,519	172,765
Total Columns	53	20 (analytical)
Duplicate Rows	0	0
Cancelled Orders Removed	7,754 rows	—
Date Range	Jan 2015 – Jan 2018	Jan 2015 – Jan 2018

Missing Values (Pre-Cleaning)

Column	Missing Count	% Missing
Product Description	180,519	100.0% — Dropped
Order Zipcode	155,679	86.2% — Dropped
Customer Last Name	8	<0.01%
All other columns	0	0.0% — Complete

Categorical Variable Distributions

Payment Type	Count	Customer Segment	Count
DEBIT	69,295 (40.1%)	Consumer	89,420 (51.8%)
TRANSFER	42,129 (24.4%)	Corporate	52,528 (30.4%)
PAYMENT	41,725 (24.2%)	Home Office	30,817 (17.8%)
CASH	19,616 (11.3%)	—	—

Shipping Mode	Orders	Order Status	Orders
Standard Class	103,153 (59.7%)	COMPLETE	59,491 (34.4%)
Second Class	33,806 (19.6%)	PENDING_PAYMENT	39,832 (23.1%)
First Class	26,513 (15.3%)	PROCESSING	21,902 (12.7%)
Same Day	9,293 (5.4%)	PENDING + others	51,540 (29.8%)

Business KPIs & Performance Dashboard

Core performance indicators derived from the supply chain dataset. These KPIs underpin all subsequent analysis.

Core Business Metrics

172,765	94,523	45.29%	54.71%
Total Orders	Late Deliveries	On-Time Rate	Late Delivery Rate

\$7.5M	\$2.1M	3 Days	3.47 Days
Total Profit	Loss from Delays	90th Pctile Delay	Avg. Processing

On-Time vs Late Deliveries & Department Delay Rates

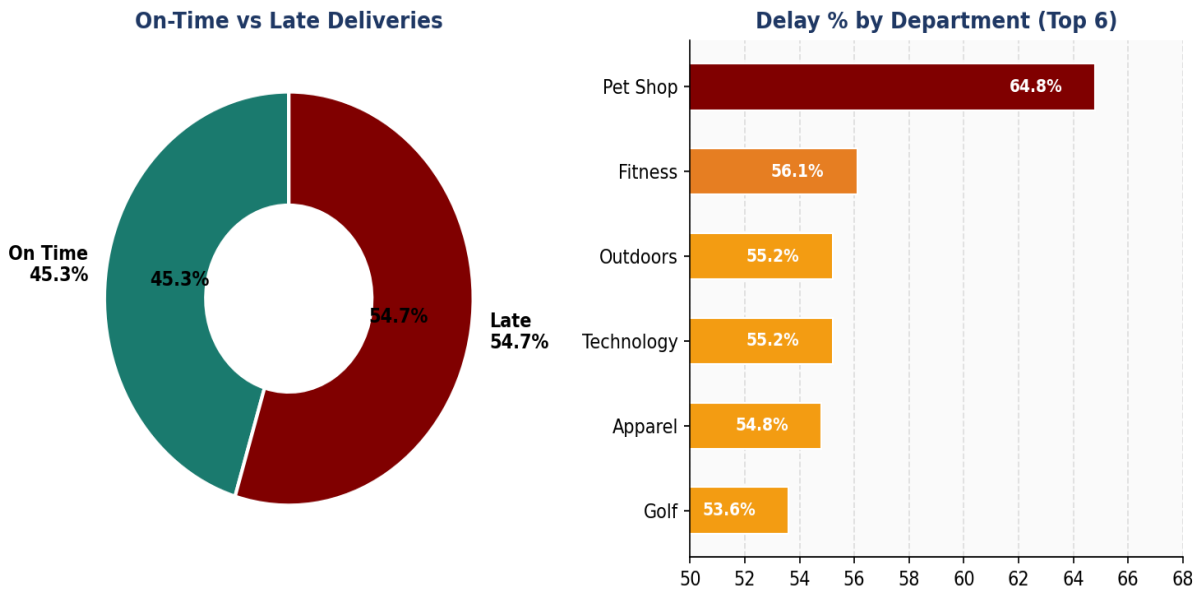


Figure 2 — Left: On-Time (45.3%) vs Late (54.7%) | Right: Delay rate by top 6 departments

Detailed KPI Breakdown

KPI	Value	Status
Total Orders Analyzed	172,765	Baseline
Late Deliveries (count)	94,523	Critical
On-Time Delivery Rate	45.29%	Below Target
Late Delivery Rate	54.71%	Critical
Total Profit (positive orders)	\$7,500,000	Positive
Profit at Risk from Delays	\$2,100,000	High Risk

90th Percentile Delay	3 days	Monitor
Mean Profit Per Order	\$22.03	Positive

Delivery Delay & Profitability Analysis

Cross-examination of delivery delay (days vs scheduled) against profitability. Negative delay = early delivery; positive = late.

Order Delay Distribution

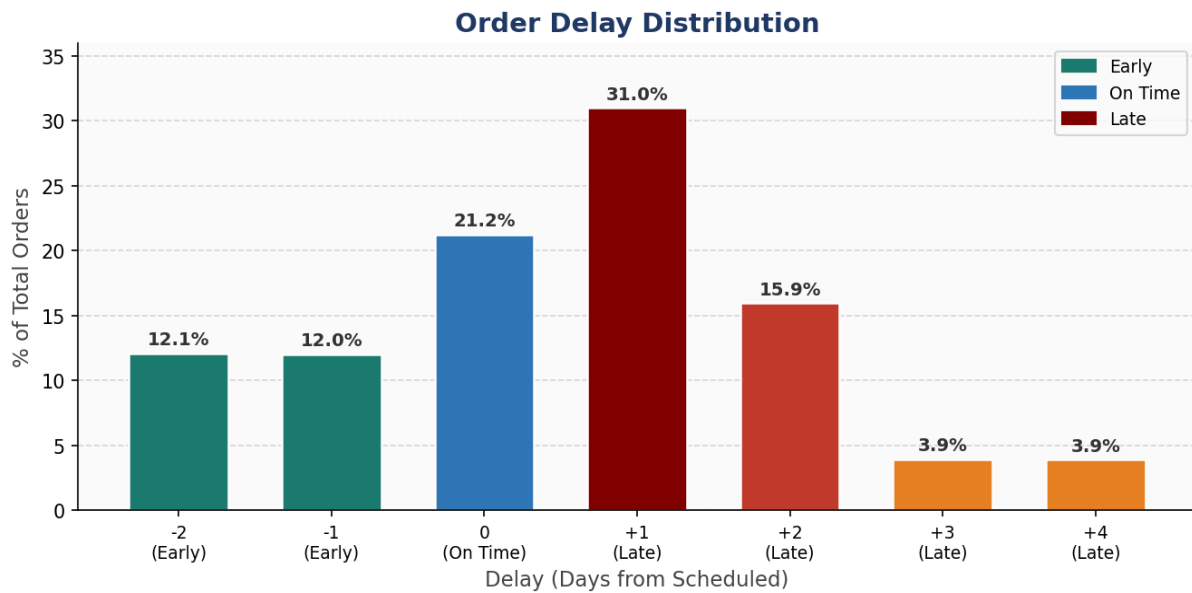


Figure 3 — 31% of orders arrive 1 day late; only ~45% arrive early or on time

Delay	Order Count	% of Orders	Classification	Flag
-2 (Early)	20,873	12.08%	Early	Good
-1 (Early)	20,719	11.99%	Early	Good
0 (On Time)	36,650	21.21%	On Time	OK
+1 (Late)	53,503	30.97%	Late	Risk
+2 (Late)	27,551	15.95%	Late	Risk
+3 (Late)	6,772	3.92%	Very Late	Critical
+4 (Late)	6,697	3.88%	Very Late	Critical

Profit Analysis by Delay Days

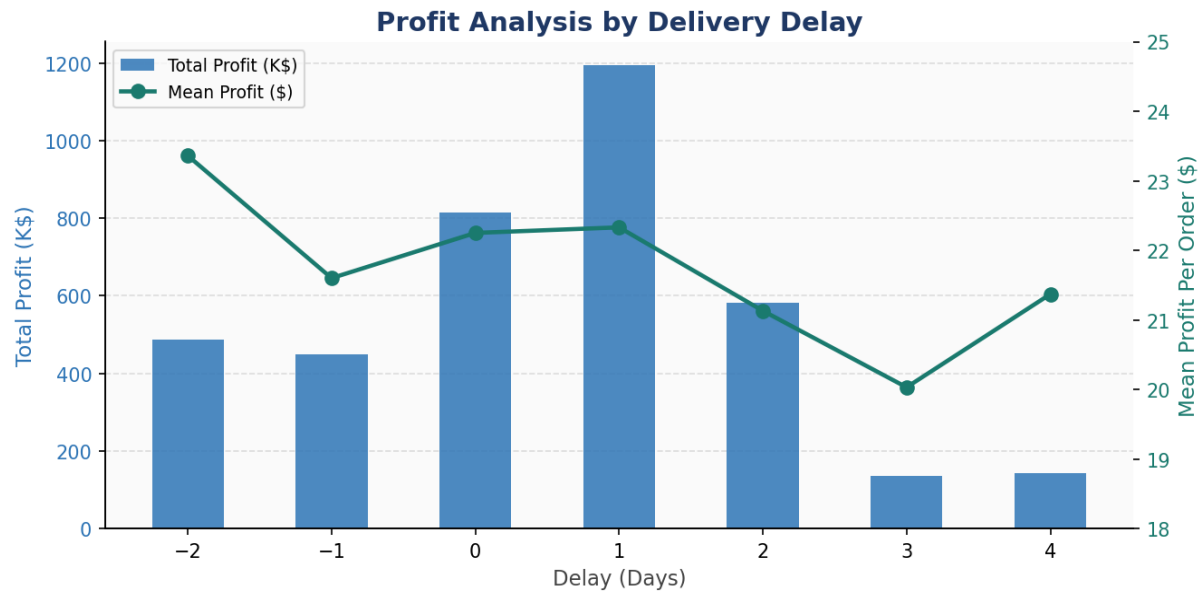


Figure 4 — Total profit (bars) vs Mean profit per order (line): profit erodes as delays worsen

Delay	Mean Profit/Order	Total Profit	Observation
-2 (Early)	\$23.36	\$487,596	Best mean — early dispatch = higher margins
-1 (Early)	\$21.60	\$447,629	Slightly below -2 day early
0 (On Time)	\$22.25	\$815,430	High volume drives strong total profit
+1 (Late)	\$22.33	\$1,194,895	Highest total due to volume — not a benefit
+2 (Late)	\$21.13	\$582,111	Declining mean as delays increase
+3 (Late)	\$20.03	\$135,653	Lowest mean — severe delays hurt margins
+4 (Late)	\$21.37	\$143,107	Slight recovery — possibly urgent high-value orders

Bottleneck Detection

Identifying operational dimensions most associated with delivery delays across all key categories.

Delay Rate by Order Region

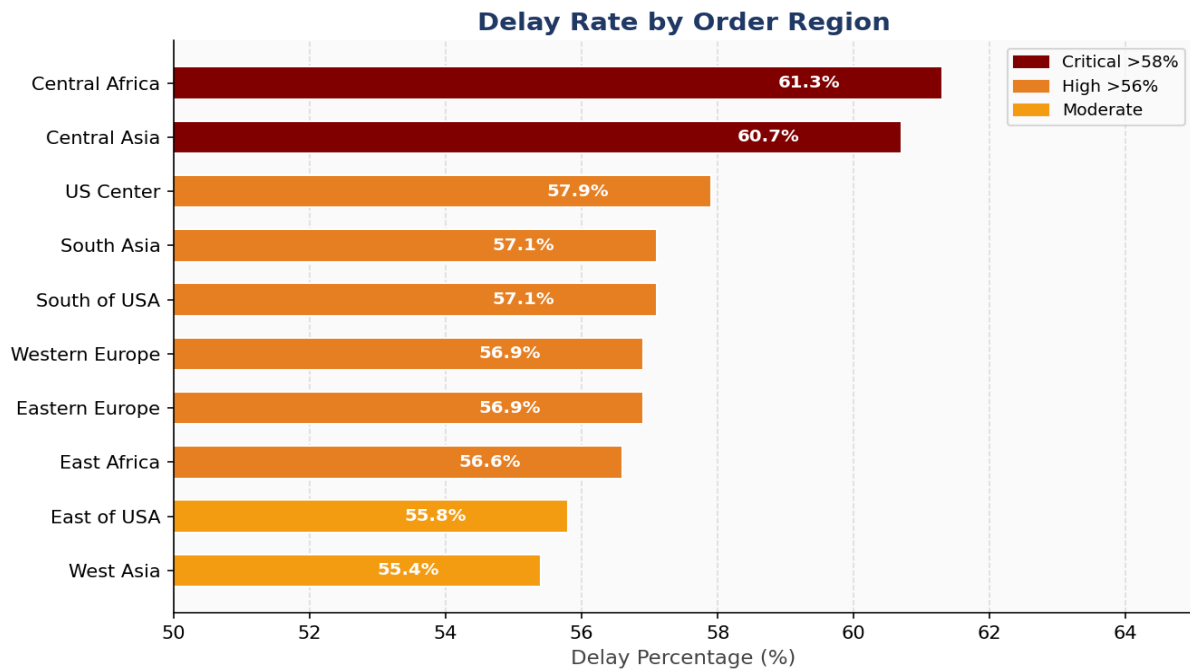


Figure 5 — Central Africa (61.3%) and Central Asia (60.7%) are the highest-risk regions globally

#	Region	Delay %	Risk Level
1	Central Africa	61.3%	Critical
2	Central Asia	60.7%	Critical
3	US Center	57.9%	High
4	South Asia	57.1%	High
5	South of USA	57.1%	High
6	Western Europe	56.9%	Moderate
7	Eastern Europe	56.9%	Moderate
8	East Africa	56.6%	Moderate
9	East of USA	55.8%	Moderate
10	West Asia	55.4%	Moderate

Delay Rate by Shipping Mode

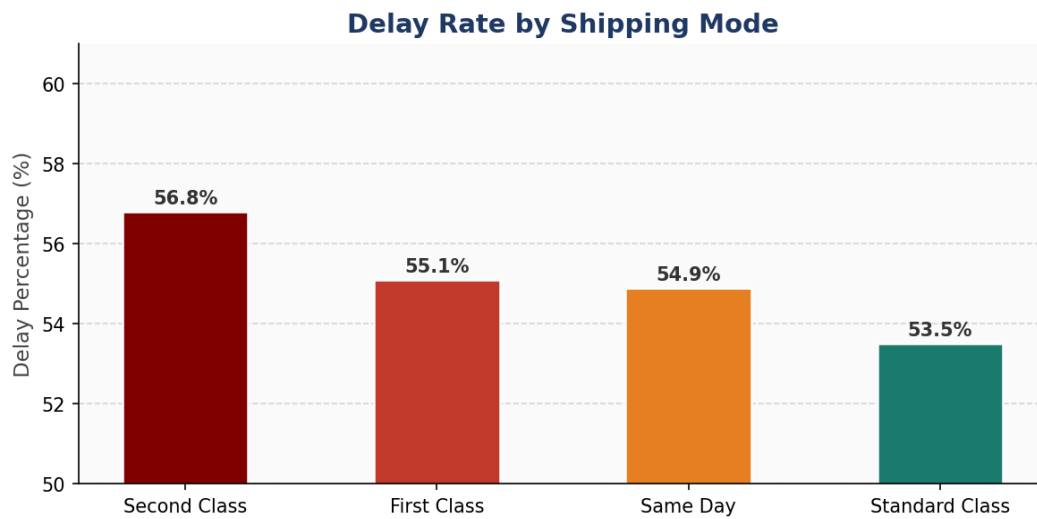


Figure 6 — Second Class has the highest delay rate; Standard Class is most reliable

Delay by Order Status & Payment Type

Order Status	Delay %	Payment Type	Delay %
ON_HOLD	60.3% (Highest)	PAYMENT	56.4%
PAYMENT_REVIEW	58.0%	TRANSFER	53.8%
PENDING_PAYMENT	56.3%	DEBIT	53.2%
PENDING	54.6%	CASH	51.9% (Lowest)
COMPLETE	53.7%	—	—

Root Cause Analysis — Central Africa

Deep-dive into Central Africa, the region with the highest overall delay rate (61.3%). Multiple driver dimensions were evaluated to isolate the top contributors to late delivery.

Top 10 Delay Drivers — Central Africa

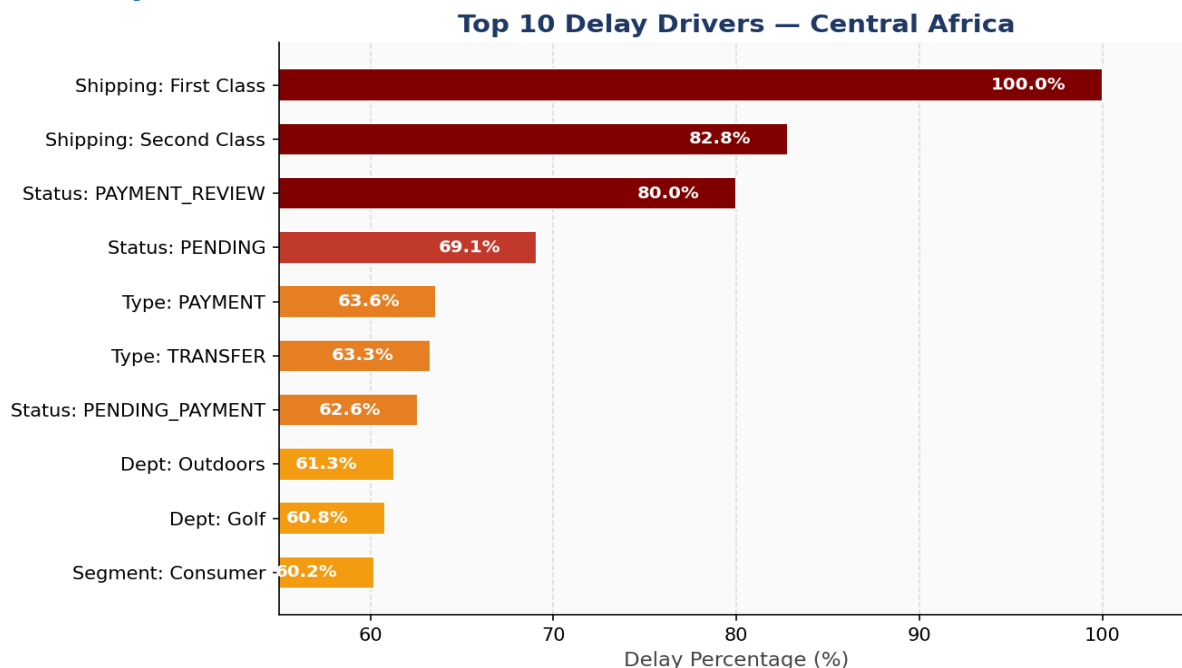


Figure 7 — First Class shipping (100% delay) and Second Class (82.8%) are the dominant failure modes

Rank	Driver Factor	Delay %	Category
1	Shipping Mode: First Class	100.0%	Shipping Mode
2	Shipping Mode: Second Class	82.8%	Shipping Mode
3	Order Status: PAYMENT_REVIEW	80.0%	Order Status
4	Order Status: PENDING	69.1%	Order Status
5	Type: PAYMENT	63.6%	Payment Type
6	Type: TRANSFER	63.3%	Payment Type
7	Order Status: PENDING_PAYMENT	62.6%	Order Status
8	Department: Outdoors	61.3%	Department
9	Department: Golf	60.8%	Department
10	Customer Segment: Consumer	60.2%	Customer Segment

Time Series Analysis

Temporal analysis reveals cyclical patterns in delay rates, enabling proactive resource allocation, staffing, and inventory planning.

Monthly Delay Rate Trend

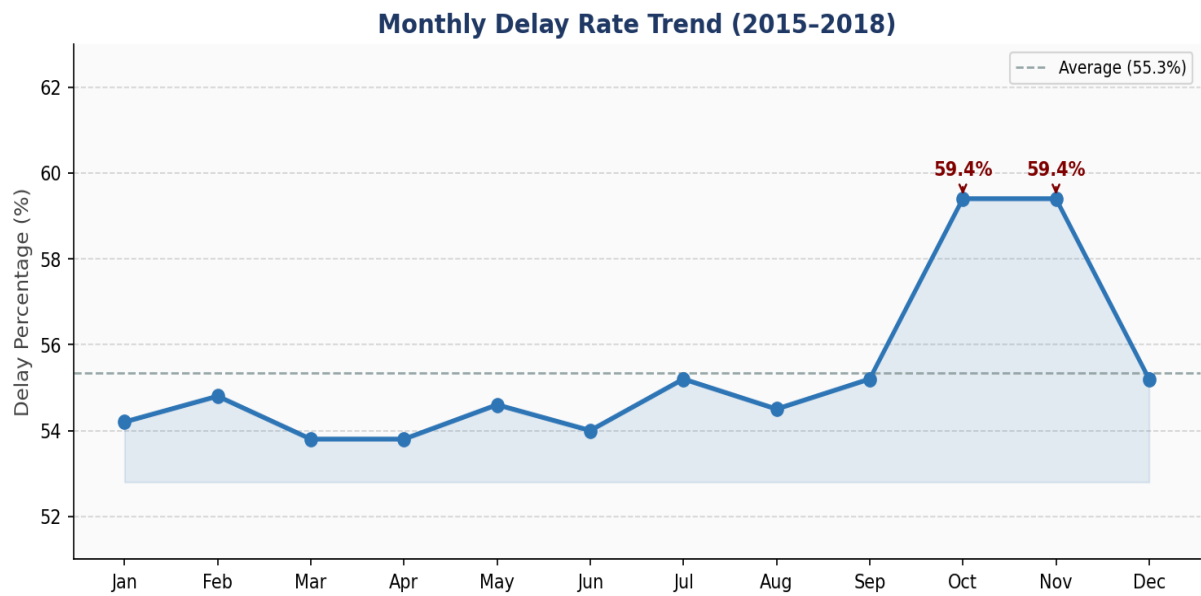


Figure 8 — Monthly delay trend: October & November peak at 59.4%; Q4 is the highest-risk quarter

Month	Delay %	Month	Delay %	Month	Delay %	Month	Delay %
Jan	54.2%	Apr	53.8%	Jul	55.2%	Oct	59.4% ▲
Feb	54.8%	May	54.6%	Aug	54.5%	Nov	59.4% ▲
Mar	53.8%	Jun	54.0%	Sep	55.2%	Dec	55.2%

Conclusions & Strategic Recommendations

Based on comprehensive analysis of 172,765 supply chain orders, the following conclusions and prioritized recommendations are presented to address operational inefficiencies and improve delivery performance.

Summary of Key Findings

#	Finding	Impact
1	54.71% of all orders delivered late — majority breach scheduled commitments	Critical
2	\$2.1M profit attributable to delayed orders — ~28% of total profit at risk	High Financial Risk
3	First Class shipping to Central Africa: 100% delay rate — zero on-time deliveries	Operational Failure
4	Pet Shop dept leads at 64.8% delay — supply chain issues in this vertical	High
5	October & November peak at 59.4% — Q4 holiday season is a systemic risk	Seasonal Risk
6	80.7% profitability rate confirms business health despite delivery challenges	Positive
7	Mean profit drops from \$23.36 to \$20.03 as delay grows from -2 to +3 days	Moderate

Strategic Recommendations

Recommendation	Target Area	Priority
Suspend or reroute First & Second Class shipments to Central Africa; renegotiate carrier contracts	Shipping / Regional	P1 — IMMEDIATE
Implement automated flags for PAYMENT_REVIEW and PENDING_PAYMENT orders to expedite fulfilment	Order Management	P1 — IMMEDIATE
Pre-position Pet Shop & Fitness inventory ahead of Q4; set seasonal surge plans	Inventory / Dept Ops	P2 — HIGH
Increase staffing for Monday orders and 10 AM–3 PM peak processing hours	Workforce Planning	P2 — HIGH
Conduct regional logistics audit for Central Africa, Central Asia, and US Center	Regional Logistics	P2 — HIGH
Deploy SLA monitoring dashboards; target 65%+ on-time delivery rate by year-end	KPI & Monitoring	P3 — MEDIUM
Investigate why CASH payments have lowest delay (51.9%) — apply learnings to other types	Payment Ops	P3 — MEDIUM

Conclusion

The DataCo supply chain is operating under significant delivery pressure, with more than half of all orders missing their scheduled dates. While the business remains profitable overall (80.7% of orders), the \$2.1M profit exposed to delay risk and complete operational failures — notably First Class shipping to Central Africa at 100%

delay — demand urgent intervention.

With targeted corrective actions in regional logistics, shipping mode optimisation, and seasonal capacity planning, the supply chain has the potential to improve its on-time rate from 45.29% to an industry-standard 70%+, unlocking substantial profit recovery and customer satisfaction gains.

End of Report — DataCo Supply Chain Analysis | May 2026